

Netflix Content Analysis

Big Data Insights Using Hadoop, PySpark & Spark SQL

CSP 554 — Big Data Technologies
Illinois Institute of Technology

Project Team

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Abstract

This report presents the findings of a comprehensive big data analysis of the Netflix titles dataset, conducted as part of the CSP 554 Big Data Technologies course at the Illinois Institute of Technology. The project employed a full end-to-end data engineering pipeline spanning Hadoop HDFS for distributed storage, Apache PySpark for large-scale data cleaning and transformation, and Spark SQL for structured analytical querying. The cleaned dataset of approximately 7,700 titles spanning the years 2008 through 2021 was subsequently visualized using Python libraries Matplotlib and Seaborn, and Tableau dashboards, revealing key insights into Netflix's content strategy. Major findings include a dominant focus on mature-audience content (TV-MA), a clear growth peak in 2019, significant geographic concentration in US and Indian productions, and Dramas as the leading genre category. These insights are presented across six analytical visualizations, each supported by structured SQL analysis.

1. Introduction

Netflix, founded in 1997 as a DVD rental service, has evolved into the world's largest subscription-based streaming platform, serving over 260 million subscribers across 190 countries. Its content library spans thousands of titles including movies, original series, documentaries, stand-up specials, and international productions. Understanding the composition and growth of this library is valuable for content strategists, data engineers, and media researchers alike.

This project undertakes a structured analysis of the Netflix titles dataset using a modern big data technology stack. The dataset, sourced from TidyTuesday's public GitHub repository, contains 7,789 records with attributes including title, type, director, cast, country, date added, release year, rating, duration, and genre listings. Through a series of milestones spanning April through May 2026, our team designed and executed a pipeline that ingests raw data into Hadoop HDFS, cleans and engineers features via PySpark, queries the structured data with Spark SQL, and produces rich visualizations for reporting.

The core research questions driving this analysis are:

- Which genres dominate the Netflix content library, and how has their representation changed over time?
- How has Netflix's total content volume grown year-over-year, and what were the peak acquisition periods?
- Which countries contribute the most titles, and what does the geographic distribution of content look like?
- What is the distribution of content ratings, and does it differ between Movies and TV Shows?
- What is the proportion of Movies vs. TV Shows, and how does duration vary across content types?

2. Dataset & Methodology

2.1 Dataset Description

The Netflix Titles dataset is a publicly available dataset maintained in the TidyTuesday repository. It contains metadata about all movies and TV shows available on Netflix as of mid-2021. The raw dataset comprises 7,789 rows and 12 columns, including: show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in (genre tags), and description.

The dataset presents several data quality challenges that necessitated a structured cleaning phase. These include approximately 2,600 null values in the director column, approximately 800 nulls in the cast and country columns, inconsistent country and genre fields (which can list multiple comma-separated values), and a date_added field stored as a human-readable string (e.g., "August 14, 2020") requiring parsing. Additionally, the duration field encodes both movie lengths (e.g., "90 min") and TV season counts (e.g., "2 Seasons") as a single mixed string.

2.2 Technology Stack

The project employed the following tools and frameworks across its five milestone phases:

- Apache Hadoop HDFS (Amazon EMR, ec2-54-144-134-116): Distributed storage for raw and processed data
- Apache Spark 3.5.6 (Amazon EMR): Distributed processing engine for PySpark and Spark SQL
- PySpark DataFrames API: Schema validation, null handling, deduplication, and feature engineering
- Spark SQL: Structured analytical queries for aggregation and trend analysis
- Python (Matplotlib 3.x, Seaborn 0.13.x): Static chart generation
- Tableau: Interactive dashboard creation for exploratory analysis

3. Data Pipeline Overview

The project was structured as a six-milestone pipeline, with each team member owning a distinct stage. The table below summarizes the milestone structure, target dates, owners, and deliverables.

MS	Date	Owner	Deliverable
M1	Apr 13	Fnu Mandakini	Literature review and project introduction submitted
M2	Apr 18	Fnu Mandakini	Hadoop HDFS setup complete; Netflix dataset ingested and schema validated
M3	Apr 22	Aditya Mandlik	PySpark data cleaning pipeline complete; cleaned data written back to HDFS (7,700 rows, 20 columns)
M4	Apr 24	Jainil Modi	All Spark SQL queries finalized; analytical results computed and exported
M5	Apr 29	Saadullah Mohammed	Python visualizations (Matplotlib/Seaborn) complete (Saadullah Mohammed); Interactive Tableau dashboards (.twbx) built and reviewed (Shifaa Nazeersaheb Nadaf)
M6	May 3	Shifaa Nazeersaheb Nadaf	Final project report submitted; presentation delivered to class

3.1 Data Ingestion (M2 — Fnu Mandakini)

The raw Netflix CSV was loaded onto an Amazon EMR cluster and ingested into the Hadoop Distributed File System (HDFS) at the path `/user/hadoop/data/netflix_titles.csv`. The ingestion step validated that the CSV was accessible by all Spark executor nodes and confirmed the 12-column schema.

3.2 PySpark Data Cleaning (M3 — Aditya Mandlik)

The cleaning pipeline, executed in PySpark on Apache Spark 3.5.6, performed six key steps: (1) explicit schema definition using `StructType`, (2) null handling — filling missing director, cast, and country values with "Unknown" and dropping the few rows with null `date_added`, `rating`, or `duration`, (3) deduplication using `dropDuplicates()` on `show_id`, (4) date parsing from string format to a proper date type, (5) feature engineering to add eight new columns, and (6) export via `coalesce(1)` to produce a single CSV at `/user/hadoop/netflix_processed/`.

Eight engineered columns were added to support Spark SQL analysis: `date_added_parsed`, `year_added`, `month_added`, `country_primary` (first country from comma-separated list), `genre_primary` (first genre from comma-separated list), `genre_count`, `duration_int` (numeric duration extracted), and `duration_type` ("minutes" for Movies, "seasons" for TV Shows). The cleaned dataset contains approximately 7,700 rows and 20 columns total.

3.3 Spark SQL Analysis (M4 — Jainil Modi)

Structured SQL queries were executed over the cleaned Spark DataFrame to compute the analytical results underlying each visualization. Queries addressed: genre distribution using GROUP BY on genre_primary; content type ratio (Movies vs. TV Shows); year-over-year counts using GROUP BY on year_added and type; top-producing countries using GROUP BY on country_primary; rating distribution stratified by type; and duration statistics by content type.

3.4 Python Visualization (M5 — Saadullah Mohammed)

Six Netflix-themed visualizations were generated using Matplotlib and Seaborn with a consistent dark color scheme mirroring the Netflix UI (background #141414, accent red #E50914, teal #00B4D8, gold #F5A623). Charts include a horizontal bar plot for genre distribution, a multi-line area chart for year-over-year trends, a vertical bar plot for geographic breakdown, a stacked bar chart for rating distribution, a donut chart plus box plot for content type and duration, and a heatmap for genre-year cross-tabulation.

3.5 Tableau Interactive Dashboards (M5 — Shifaa Nazeersaheb Nadaf)

Complementing the static Python visualizations, Shifaa Nazeersaheb Nadaf developed a suite of interactive Tableau dashboards that enable dynamic, filter-driven exploration of the Netflix dataset. Unlike static charts, the Tableau dashboards allow non-technical stakeholders including course instructors and project reviewers to drill down into findings without writing a single line of code, making the project's analytical results broadly accessible and presentation-ready.

Four interactive dashboards were created in Tableau Desktop, each connected to the cleaned netflix_cleaned.csv output from M3:

- **Genre & Content Type Explorer:** An interactive bar chart allowing users to filter by content type (Movie / TV Show) and dynamically re-rank genre categories. Filters update all charts on the dashboard simultaneously via Tableau's action-filter mechanism.
- **Geographic Content Map:** A choropleth world map visualizing title counts by country_primary, color-coded by production volume. Users can hover over any country to see its exact title count and top genre, and click to filter all other panels to that country's content profile.
- **Content Growth Timeline:** A dual-axis line chart showing year-over-year additions for Movies and TV Shows from 2008 to 2021. Users can filter by rating category (TV-MA, TV-14, PG, etc.) to observe how Netflix's content strategy within each rating tier evolved over time.
- **Rating & Audience Distribution:** A stacked bar dashboard breaking down content ratings (TV-MA, TV-14, R, PG-13, etc.) split by Movie vs. TV Show, with a companion treemap showing rating share proportions. Interactive filters allow selection by year range and country to see how rating composition shifts across markets and time periods.

All four dashboards were built with a unified dark theme consistent with Netflix's visual identity, using a black background (#141414) and red accent (#E50914). Each dashboard was reviewed by the team during M5 to validate that the interactive outputs matched the Spark SQL analytical results produced by Jainil Modi in M4. The dashboards were packaged as a Tableau Workbook (.twbx) for offline presentation and submitted alongside the final report.

4. Key Findings

The following sub-sections present the six major analytical findings derived from the Spark SQL queries and visualized by the Python pipeline. A summary of dataset metrics is provided in the table below.

Metric	Value
Raw Input Rows	7,789 rows
Cleaned Output Rows	~7,700 rows (nulls & duplicates removed)
Original Columns	12
Engineered Columns Added	8 (year_added, month_added, country_primary, genre_primary, etc.)
Total Output Columns	20
Total Distinct Titles	7,631
Movies	5,268 (69%)
TV Shows	2,363 (31%)
Years Covered	2008 – 2021
Peak Content Year	2019 (2,126 titles added)

4.1 Genre Distribution

The genre analysis, based on the genre_primary column extracted from each title's listed_in tags, reveals that Dramas are the single most prevalent content category on Netflix with 1,373 titles a margin of 323 titles over the second-place Comedies (1,050). Documentaries (730) and Action & Adventure (717) rank third and fourth, reflecting strong viewer demand for both informational and high-energy content. International TV Shows (687) rank fifth, underscoring Netflix's significant investment in global content acquisition. Children & Family Movies (492), Crime TV Shows (365), and Kids' TV (351) form the mid-tier, while Stand-Up Comedy (293), Horror Movies (242), British TV Shows (227), and Docuseries (190) complete the top 12.

The dominance of Dramas aligns with industry-wide data indicating that emotionally engaging narrative content drives higher subscriber retention metrics. The presence of both "Documentaries" and "Docuseries" as distinct categories suggests an opportunity for genre taxonomy consolidation in future analyses.

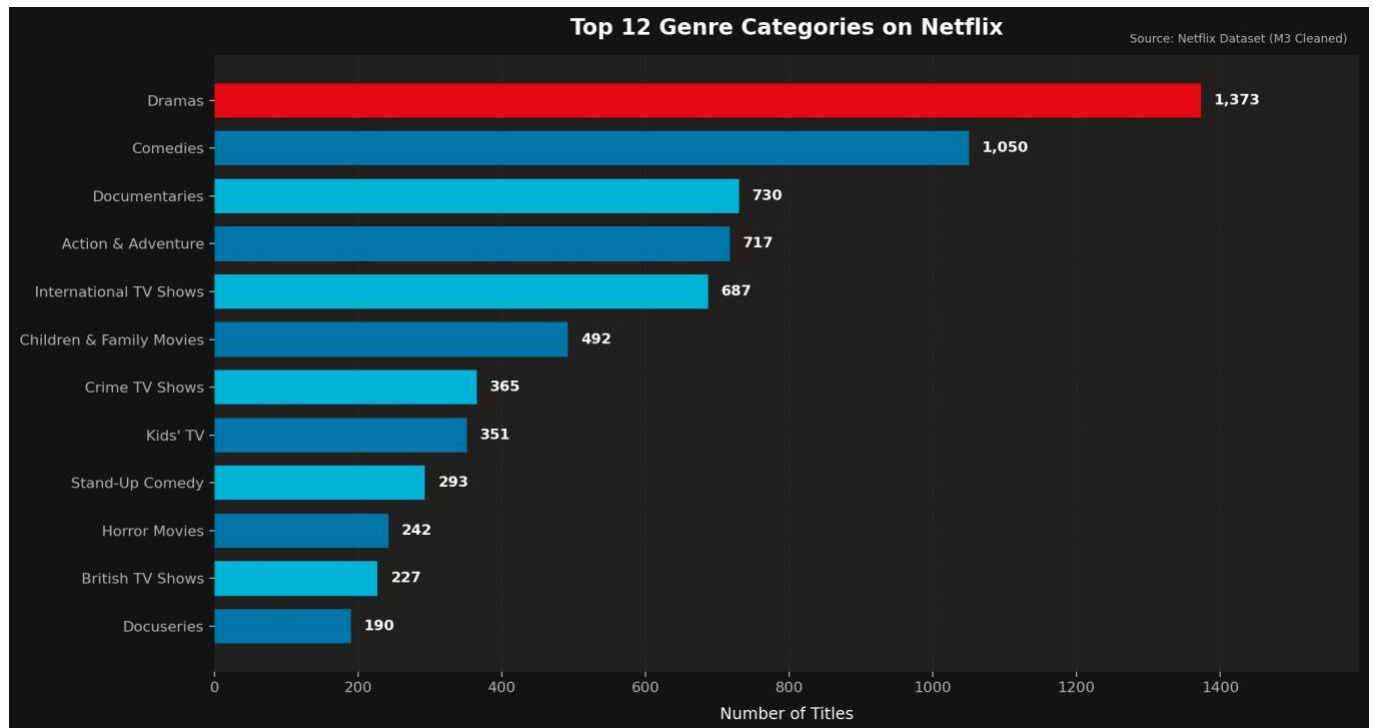


Figure 1: Top 12 Genre Categories on Netflix by Number of Titles

4.2 Year-over-Year Content Growth

The content growth analysis covers the period from 2015 to 2021, focusing on the years in which Netflix aggressively expanded its library. Total titles added grew from just 82 in 2015 to a peak of 2,126 in 2019, representing a 25-fold increase over four years. Both Movies and TV Shows contributed to this growth, with Movies consistently comprising the larger share. From 2019 onward, total additions declined — dropping to approximately 1,982 in 2020 and falling sharply to around 120 in 2021, likely reflecting both dataset coverage limitations (the dataset was assembled in mid-2021) and potential pandemic-related production slowdowns.

The 2019 peak is consistent with Netflix's widely-reported content investment strategy during that period, in which the company dramatically increased its budget for original productions and global licensing. The relatively smaller TV Show growth compared to Movies suggests that original series, while high-profile, represent a smaller share of total catalog volume.

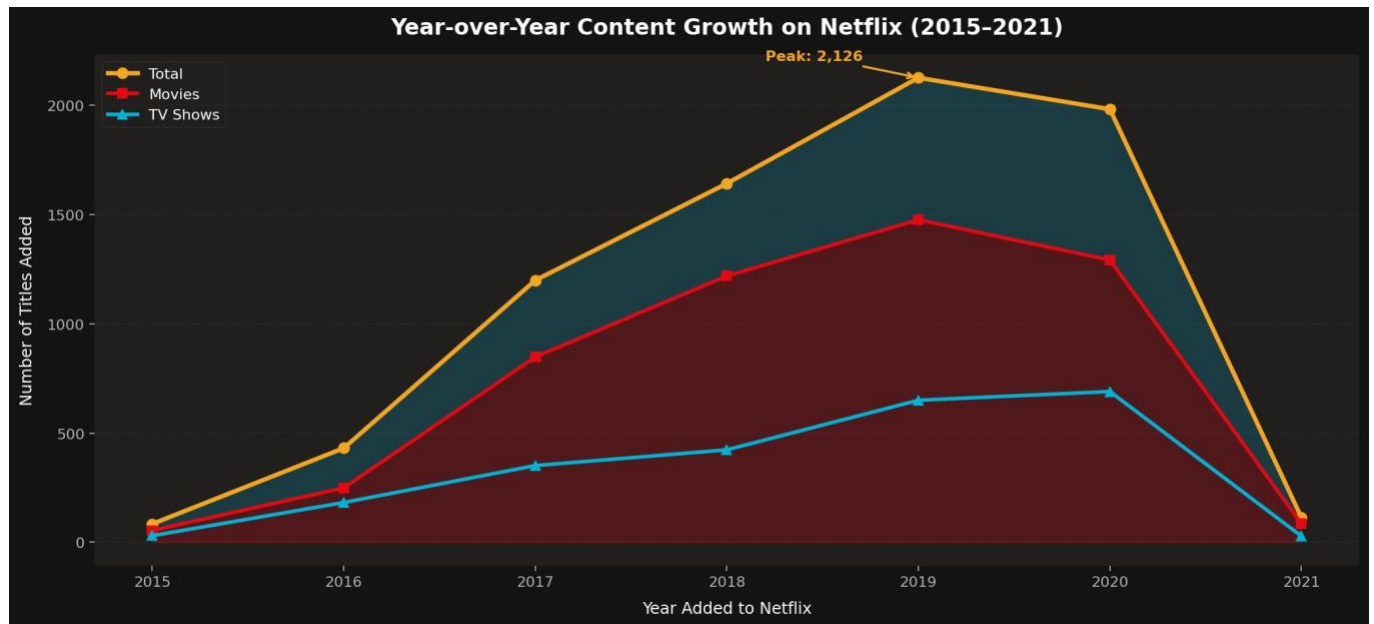


Figure 2: Year-over-Year Content Growth on Netflix (2015–2021)

4.3 Geographic Breakdown

The geographic analysis reveals a pronounced concentration of Netflix content originating from the United States, which accounts for 2,795 titles nearly three times the count of the second-largest contributor, India (949 titles). The United Kingdom ranks third at 568 titles, followed by Canada (254) and Japan (230). France (195), South Korea (194), Spain (163), Mexico (122), Turkey (106), Egypt (105), and Australia (103) round out the top 12.

The strong representation of Indian content reflects Netflix's strategic investment in Bollywood and regional Indian-language productions to capture one of the world's largest streaming markets. South Korea's position (194 titles) is notable given its population size, and reflects the global success of Korean drama content a trend that would accelerate significantly after the dataset's coverage period with titles such as *Squid Game*. Approximately 800 rows with unknown country values were excluded from this analysis.

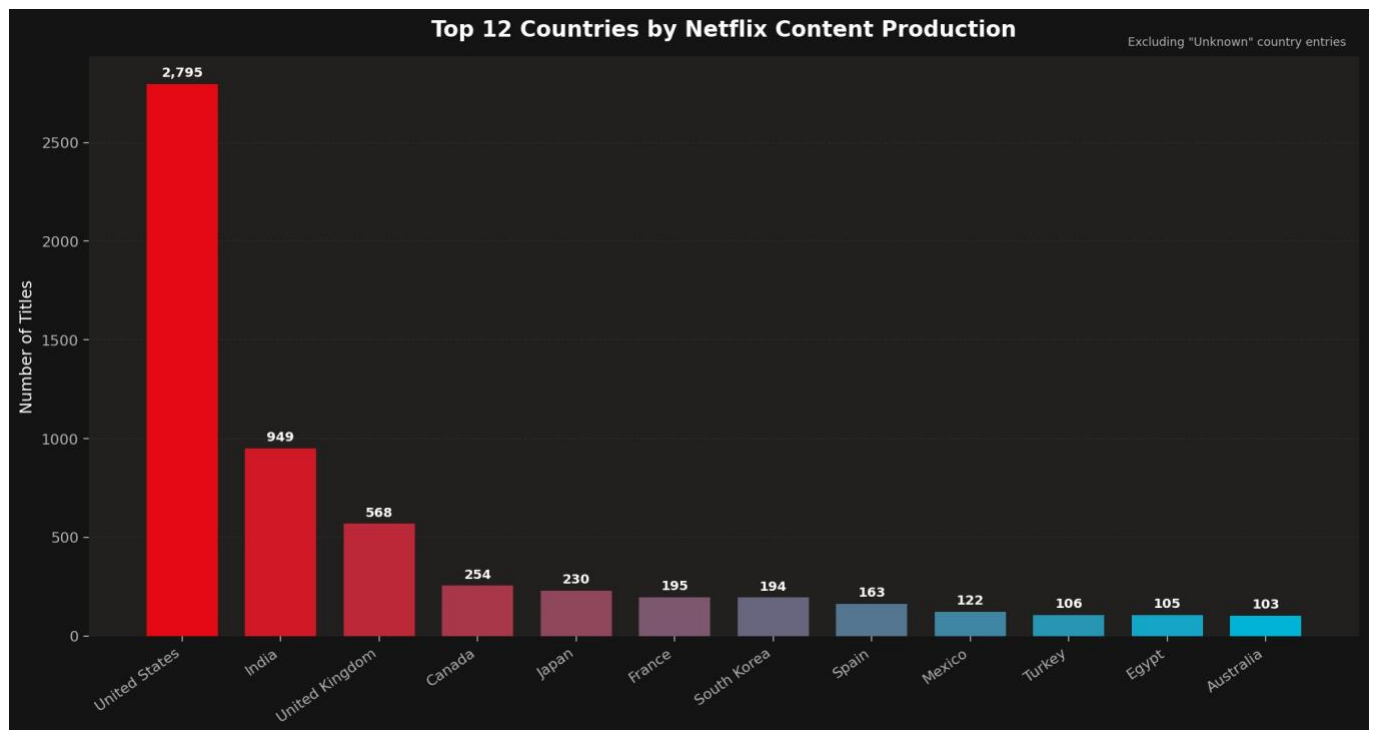


Figure 3: Top 12 Countries by Netflix Content Production (Excluding Unknown Entries)

4.4 Rating Distribution

The rating distribution analysis reveals a clear skew toward mature-audience content. TV-MA (Mature Audiences) dominates with 2,797 total titles, composed of approximately 1,800 Movies and nearly 1,000 TV Shows. TV-14 is second with 1,900 titles. Together, TV-MA and TV-14 account for approximately 62% of the entire Netflix catalog confirming that the platform's content strategy is predominantly targeted at adult and teenage viewers.

Content in the Kids/Family zone (G, TV-G, PG, TV-Y, TV-Y7, TV-Y7-FV) totals approximately 969 titles. The R-rated Movie segment contributes 658 titles, predominantly in the Movie category. The relatively small NR (Not Rated, 80 titles) and UR (Unrated) categories likely contain older catalog titles and international content for which US rating data was unavailable. The absence of NC-17 titles at meaningful scale indicates Netflix does not pursue that content segment.

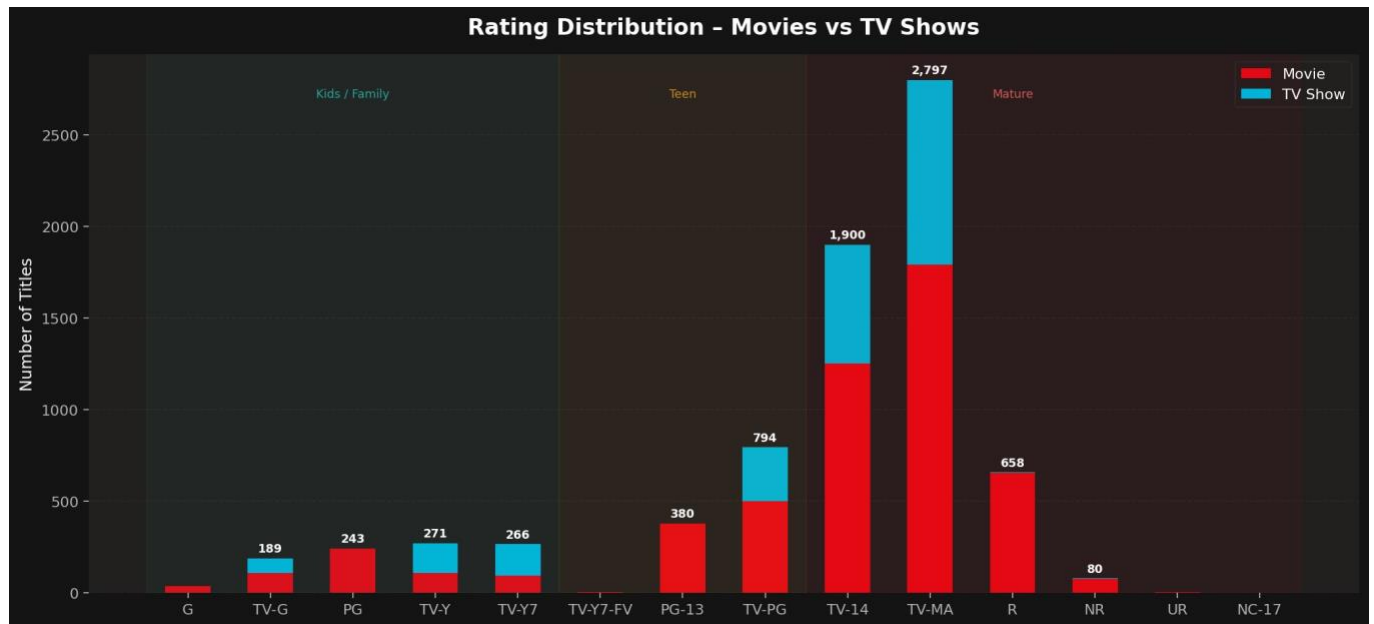


Figure 4: Rating Distribution — Movies vs TV Shows Stacked by Content Type

4.5 Content Type Split and Duration

Of the 7,631 titles in the cleaned dataset, 69% are Movies (approximately 5,268 titles) and 31% are TV Shows (approximately 2,363 titles). This 2:1 ratio underscores that while Netflix's original TV series receive the most press coverage, the platform's catalog volume is substantially movie-dominated.

The box plot analysis of duration reveals that the median Movie runtime is approximately 100 minutes, with an interquartile range of roughly 85 to 115 minutes. Outliers at both extremes exist ranging from very short features (under 40 minutes) to extremely long films (over 200 minutes). TV Shows have a median duration of 1 season, with the majority of series in the dataset having 1 to 3 seasons. The presence of outliers above 10 seasons reflects long-running legacy series licensed from broadcast television.

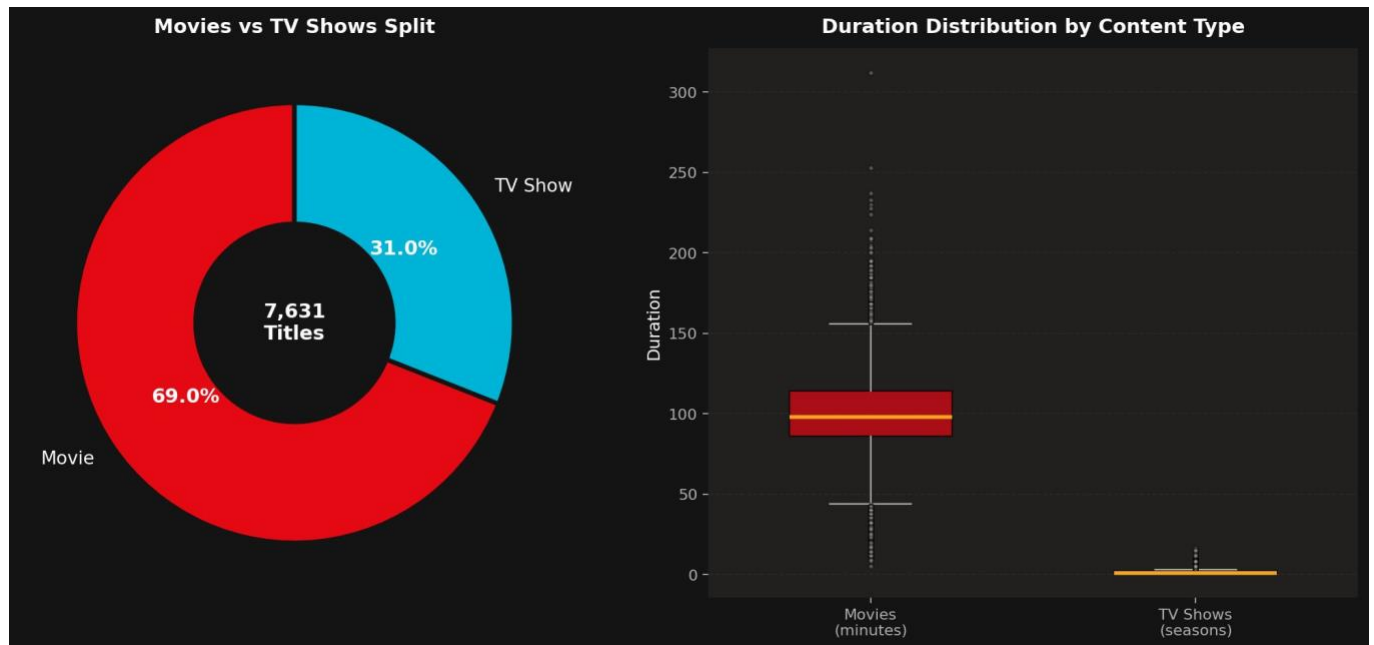


Figure 5: Movies vs TV Shows Split (Donut) and Duration Distribution by Content Type (Box Plot)

4.6 Genre-Year Heatmap

The genre-year heatmap provides a cross-tabulation of the top 10 genres by year (2016–2021), revealing temporal patterns in Netflix's content strategy. Dramas show the highest absolute cell values across all years, peaking at 374 titles added in 2019 and remaining high in 2018 (352) and 2020 (354). Comedies follow a similar pattern, peaking at 305 titles in 2019 and 300 in 2020.

Documentaries display a distinct pattern relatively flat across 2017–2019 (184, 159, 184) before declining in 2020 (107), suggesting a different acquisition strategy compared to narrative genres. Action & Adventure titles show steady growth from 20 in 2016 to a peak of 221 in 2019, then declining. International TV Shows grew significantly in 2019 (204) and 2020 (168), consistent with Netflix's global expansion. Children & Family Movies showed a notable upward trend through 2020 (156), while Stand-Up Comedy peaked in 2017 (60) and 2018 (77) before tapering. The near-zero counts in 2021 across all genres reflect the mid-year cutoff of the dataset rather than actual content decline.

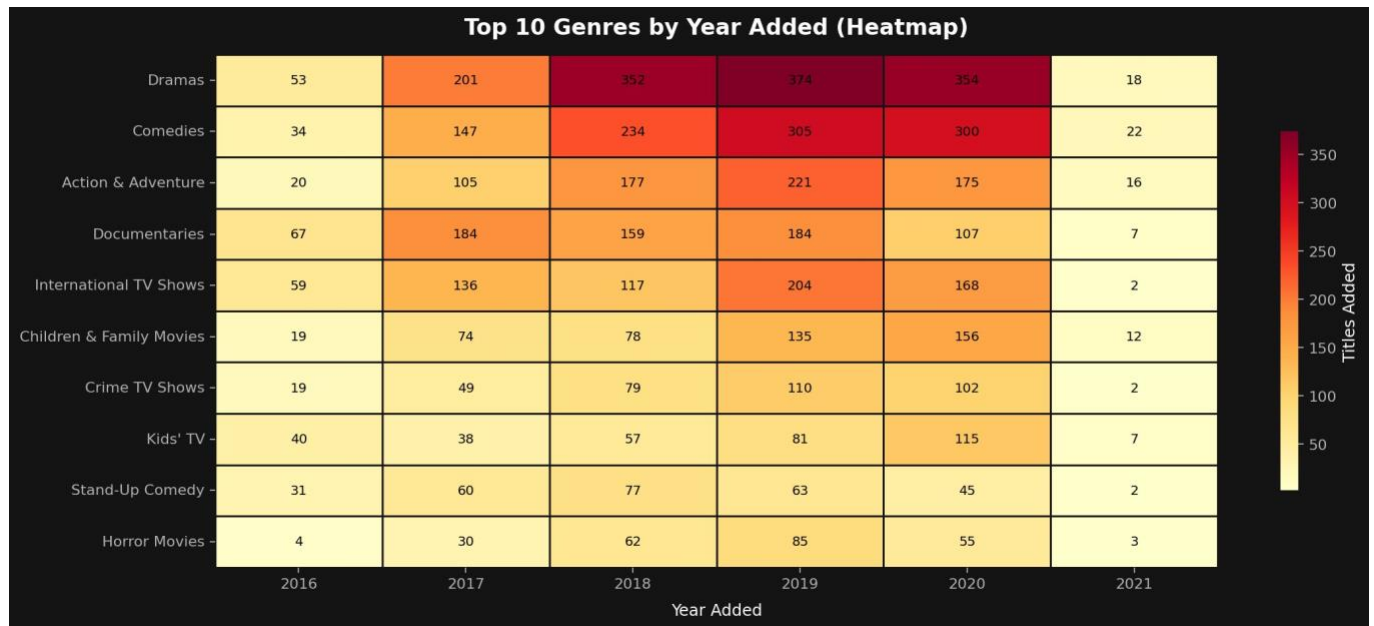


Figure 6: Top 10 Genres by Year Added (Heatmap, 2016–2021)

5. Conclusions & Recommendations

This project successfully demonstrated an end-to-end big data pipeline from raw CSV ingestion through Hadoop HDFS to structured Spark SQL analysis and rich Python visualizations. The Netflix titles dataset, while modest in volume relative to true enterprise-scale datasets, provided a realistic context in which to apply distributed computing tools and explore meaningful analytical questions about a globally recognized content platform.

Five key conclusions emerge from the analysis:

- Netflix's content library is drama-dominated, with Dramas (1,373 titles) and Comedies (1,050) forming the backbone of the catalog. This finding has practical implications for content recommendation systems and catalog management.
- Content acquisition peaked sharply in 2019 at 2,126 titles, driven primarily by Movie additions. The subsequent decline reflects both dataset coverage limitations and potential market consolidation pressure.
- The United States accounts for 36% of identified content production (2,795 of ~7,700 titles), with India as a strong second (12%). This geographic concentration presents both a market risk and an opportunity for further global diversification.
- The platform's content skews heavily toward mature audiences: TV-MA and TV-14 titles together represent over 60% of the catalog. This is consistent with Netflix's positioning as a premium adult entertainment service.
- The 69:31 Movie-to-TV-Show ratio reveals that while original series dominate Netflix's marketing, movies form the quantitative backbone of its catalog.

Recommendations for future work include:

- Expanding the analysis to include viewer rating data (e.g., from IMDb or Rotten Tomatoes APIs) to correlate content volume with quality perception.
- Integrating subscriber growth data to assess whether content addition volume correlates with subscriber acquisition or retention.
- Applying machine learning clustering (e.g., K-Means in Spark MLlib) to identify natural content groupings beyond the given genre taxonomy.
- Building a real-time streaming pipeline using Apache Kafka and Spark Structured Streaming to track new content additions as they are published.

6. References

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